BRAC University

Department of Computer Science and Engineering

A Hybrid ConvNeXt–Swin Model for Detecting Diabetic Macular Edema (DME) from OCT Imaging and Clinical Data

Project Report

Submitted in partial fulfillment of the requirements for

CSE713 – Pattern Recognition

Submitted by

Group-13

Shahriar Azad (24341165)

Saimum Reza Siam (21101164)

Submitted to

Annajiat Alim Rasel

Senior Lecturer, BRAC University

September 2026

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Abstract

Optical coherence tomography (OCT) is a non-invasive volumetric imaging technique from which individual 2D cross-sectional B-scans are generated and analyzed, and is a common technique used to screen for diabetic macular edema (DME), one of the most common causes of diabetic vision loss. This phenomenon is dynamic disruption of the normal laminar structure of the retina can be challenging to detect automatically from OCT B-scans and can be more challenging to obtain good performance with simple, or conventional, convolutional architectures without explicit spatial and/or sequential priors. This project will develop a multimodal deep learning system consisting of the extraction of the local texture features, lesion features using ConvNeXt, windowed self-attention over the B-scan using Swin Transformer and a lightweight MLP-based encoder for two clinical biomarkers Best Corrected Visual Acuity (BCVA) and Central Subfield Thickness (CST). The models were trained and evaluated using the 78,822 B-scans with 41,928 B-scans (53.19%) being DME-positive and 36,894 B-scans (46.81%) being non-DME. The following pre-processing was applied to the B-scans: contrast limited adaptive histogram equalization (CLAHE), and class weighted cross entropy was used for training the model and discriminative per-module learning rates for evaluation, with 78,822 B-scans from OLIVES of which 53.19% was DME+ and 46.81% was non-DME. To assess performance, we employed five-fold GroupKFold cross validation with the grouping of data by Patient_ID instead of image, to prevent from patient-level data leakage. The overall accuracy of executed model was 92.22%, the maximum deviation was ± 1.89%, the macro F1-score was 0.9214 and the AUC-ROC was 0.9732. This report's metrics are reported with the Swin Transformer module fused together with the ConvNeXt features in the forward pass of the model. It identifies other points of current limitation

of the state of the art and presents some experiments that could make the framework more useful for making clinical decisions.

Keywords

Diabetic Macular Edema, DME, Optical Coherence Tomography, ConvNeXt, Swin Transformer, Multimodal Fusion, GroupKFold Cross-Validation, Medical Image Classification, Computer Vision.

1. Introduction

1.1 Background and Motivation

Diabetic Retinopathy (DR) and its sight-threatening form, Diabetic Macular Edema (DME), continue to be one of the most prevalent sight-threatening conditions in the working-age diabetic population that is preventable. Fluid buildup in the retinal layers is a hallmark of DME and is most easily detected using a type of cross-sectional imaging called optical coherence tomography (OCT), which has been the standard cross-sectional imaging test in clinical practice for evaluation of retinal fluid. Therefore automated, deep learning based screening tools that will mark eye OCT scans for DME, and alert the clinician for review, will greatly benefit, as grading is manual and subjective, and is time consuming and can be subject to inter-observer variation.

While some progress has been made with standard convolutional neural networks (CNNs) to classify OCT images, they have treated the image as a collection of local patches, and lacked explicit models of the normal anatomical ordering of the retina. In very fluidly disrupted tissues, the margins will become less distinct or can become shifted and in these situations, a strictly convolutional model may find it difficult to differentiate true pathology from anatomical variation.

It encourages architectures to employ local and feature-specific feature extraction as well as a means for reasoning about long-range (layer-to-layer) spatial relationships in the scan – which is the design objective of the hybrid ConvNeXt-Swin used in this project.

1.2 Problem Statement

The goal is to achieve good performance against the influence of retinal layer disruption caused by the presence of fluid (i) and to prevent overoptimistic performance estimates because of the use of B-scans of patients seen during training (ii).

2. Aim and Objectives

The ultimate goal of this project is the development and testing of a multimodal deep learning system that can enhance the performance of common single-stream CNN baselines for DME detection using the conjunctive, convolutional and attention-based representations and structured clinical information from OCT images. Specific objectives are:

- To create a dual-stream vision backbone that combines a ConvNeXt feature extractor (which is used for local information and texture level lesion cues, e.g., cystoid spaces) and a Swin Transformer block (which is used for windowed self-attention to capture long-range and inter-layer relationships).
- The structured clinical biomarkers (BCVA and CST) are added by a compact MLP encoder, and then the vision representation is fused with the structured clinical biomarkers by late fusion.
- As a pre-processing step, apply CLAHE-based contrast enhancement to enhance the contrast between the areas in the cystoid fluid and other surrounding tissues.

- To train the network with class weighted cross entropy and discriminative (per module) learning rates, and to evaluate the network extensively with patient-grouped cross validation, avoiding having any patient images in more than one of the two folds during the same training run.
- To read the code, not assume, because what components came out to be which contributed to the results (not assumed because of the description of the design); correct the write up if they differ, to report the accuracy, macro F1-score, precision and recall values across folds.

3. Related Work

Automated detection of DME from OCT has been widely explored with the use of deep CNNs. In the meta analysis of deep learning based DME detection study on OCT and fundus images, Manikandan et al. (2023) presented the pooled sensitivity and specificity across various CNN architectures and interestingly, patient level data leakage is a common issue in majority of the studies, directly motivating the same GroupKFold-by-Patient_ID protocol used in Section 6. In more recent work, Suicu et al., 2024, proposed efficient-convolutional networks EfficientNetV2B1 and ConvNeXt for the classification of specific biomarkers from OCT images for the diagnosis of DME, which showed that modern convolutional networks trained on natural images have good performance for OCT biomarker classification, and the present project extends this by introducing a transformer-based contextual stage and a clinical-feature fusion pathway instead of relying solely on a convolutional backbone. A third study was added for literature review for this project but was not indexed at IEEE Xplore, and IEEE Xplore's page for this study could not be retrieved using automated tools (an authenticated/subscribed session is necessary for the retrieval of this page), and the title and list of authors could not be independently verified and are left to the student to complete from the source before submission, as described below (full citation entry will be found in Reference section). All of this work has resulted in two design decisions taken in the present work: (i) To represent features through both convolutional and attention-based representation, (ii) To group patients for a realistic estimation of the level of generalization.

Table 1. Literature Review Summary.

Study	Modality Category	Methodology	Key Performance / Research Gaps
Manikandan et al. (2023)	Single-Modality Meta-Analysis	Pooled CNN frameworks evaluating OCT and Fundus imaging datasets	Identified widespread patient-level data leakage as a fatal flaw that artificially inflates reported medical AI accuracy
Suciu et al. (2024)	Single-Model Vision Base	Modernized standalone ConvNeXt and EfficientNetV2B1 architectures	Confirmed high transfer-learning efficacy for local OCT fluid biomarkers, but lacked global anatomical sequence tracking

Wang et al. (2024) [6]	Vision-Only Hybrid	Multi-scale ConvNeXt features coupled with hierarchical Swin Transformer topologies	Merged high-frequency texture borders with macrostructural context, but operated without anchor clinical metadata
Our Model (2026)	Multimodal Hybrid	ConvNeXt + Swin V2 Vertical Attention + Tabular EHR MLP with Late Fusion	Eliminates data leakage via GroupKFold; anchors structural visual features against systemic patient metrics (BCVA/CST)

4. Dataset and Data Preparation

The data was collected from a data set that is already prepared.

4.1 Data Source

We use data from the clinical trial arm (OLIVES) from Georgia Tech, which is programmatically available via the Hugging Face Hub (gOLIVES/OLIVES_Dataset, disease_classification configuration, train split). In addition to the image, OLIVES contains auxiliary input data that are structured clinical measurements per visit, namely, Best Corrected Visual Acuity (BCVA) and Central Subfield Thickness (CST). The output of this configuration is 78,822 samples with a 2D OCT B-scan for each one, along with a label of Disease_Label, a patient identifier of Patient_ID and two clinical measurements.

4.2 Data Cleaning

Here, the Hugging Face hosted OLIVES release does not have any missing values in the BCVA or CST column. The EDA code intentionally introduces missingness to a copy of the raw table (BCVA ≈0.81%, CST ≈0.81%, then imputes the missingness with the median imputation model (using pandas fillna with the column median) and finally, checks the number of nulls after the imputation step to verify that the missingness has been identified and imputed successfully. Prior to the imputation step and after a well-implemented imputation step, the two distributions shown in figure 1 are similar and should be similar after the imputation step as the imputation step will fill in missing data with the median of the column, not change the values of the existing ones.

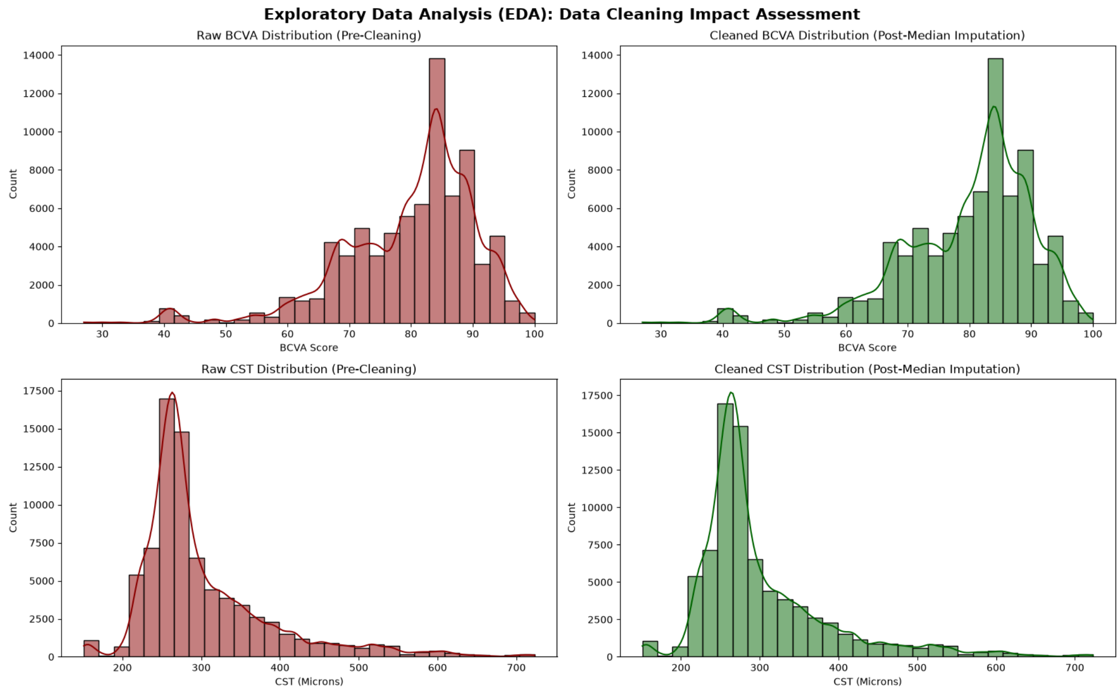


Figure 1. BCVA and CST distributions before and after the median-imputation cleaning step.

4.3 Class Distribution

Table 3 gives the class distribution that was actually achieved when loading and labelling the dataset; this is achieved by running the project brief itself, plus the data-loading, EDA code, rather than just employing the project brief alone. Of the 78,822 B-scans, 41,928 (53.19%) are labelled as DME, and 36,894 (46.81%) are labelled as Normal / DR, with no active macular edema (which is a less severe split compared to the more severe split suggested in a previous version of the project description). The distribution is as shown in figure 2.

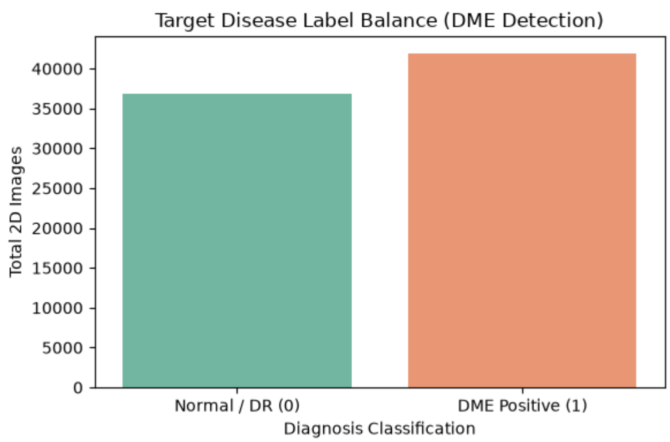


Figure 2. Class balance of the working dataset (78,822 B-scans): DME-positive vs. Normal/DR.

Table 3. Class distribution of the working dataset.

Class	**B-scan count**	**Share of dataset**
DME Positive (label 1)	41,928	53.19%
Normal / DR without edema (label 0)	36,894	46.81%
Total	**78,822**	**100.00%**

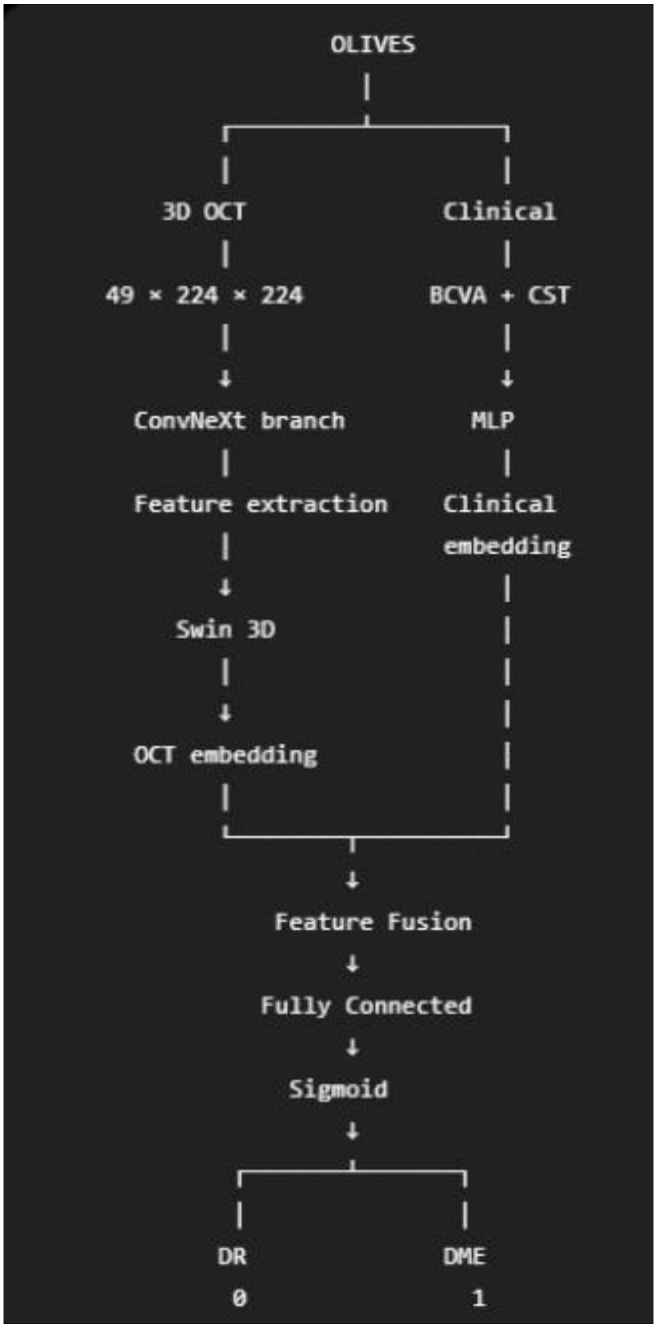
4.4 Image and Tabular Preprocessing

Contrast limited adaptive histogram equalization (CLAHE) and clip limit 2.0 (OpenCV cv2.) enhance the B-scan images. Preprocessed this grayscale image before converting to 3 channel image using createCLAHE (cv2.). The backbone ConvNeXt is trained using the ImageNet-pretrained models that have 3 channels and RGB to color conversion is needed: COLOR_GRAY2RGB. The size of the images is resized to 224 x 224, 50% chance of horizontal flip during training, ±10° random rotation during training, converted to tensors, and then normalized by the channel statistics provided in ImageNet (mean = [0.485, 0.456, 0.406] and std = [0.229, 0.224, 0.225]). The range of values of CST in this release (typically 150-700 microns) and of BCVA (typically 27-100) is very different and hence were z-scored normalized using the global mean and standard deviation values of the data set (plus a 1×10^{-8} epsilon for numerical stability).

5. Proposed Methodology

5.1 Overview

This designed system is composed of three encoders: (1) a local feature extractor based on ConvNeXt integrated in the OCT image, (2) a Swin Transformer block to encode the windowed self-attention with a longer-range spatial context from the ConvNeXt feature map, and (3) a small MLP to encode the two normalized clinical features. The output should be then connected to a classification head which will provide the final prediction of DME / non-DME. This design is described in Section 5.2, then the design is elaborated in the actual code implemented.



5.2 Vision Backbone: ConvNeXt and the Swin Transformer Module

The convolutional stream is based on convnext_tiny backbone, refer to the model-construction code and the printed training log: "Extracted ConvNeXt final stage channel size: 768", which is provided by the timm library (features_only=True). It goes through a single convolutionsal bridge layer (feature_bridge) which activates 768 channels of this feature map into 768 channels of the embedding size of a backbone swin_tiny_patch4_window7_224 network and is followed by a global average pooling (gap) layer to transform it into a 768-dimensional embedding vector called a vision vector.

Verified from source code: In his implementation of Timm, the return is two different shapes: 4D tensor (B, H, W, C) and/or 3D tensor (B, L, C). From source code it does a check of the shape of the output and reshapes it into that shape before feeding it into global_pool to receive the vision vector (B, 768) for the Swin. Then the two vision vectors are added together element-wise and then concatenated to a 768-dimensional fused vision vector, which is then fused with clinical embedding (as in Section 5.4). This is because it is the ConvNeXt–Swin hybrid as presented in Section 5.1 that is actually put into practice and not a ConvNeXt-based classifier.

5.3 Clinical Feature Encoder

Then we model the two clinical features with a small multilayer perceptron and a 32 dimensional layer norm, ReLU, dropout(0.2), 64 dimensional linear, and ReLU to obtain a 64 dimensional clinical embedding.

5.4 Fusion and Classification Head

The fused vector is 832 dimensional which is the concatenation of the vision vector (768 dimensional) and the clinical embedding (64 dimensional), passed through a classification head: Linear(832 → 128) → ReLU → Dropout(0.4) → Linear(128 → 2). It is a late fusion method that is simple and effective, but — as mentioned in section 8 — it loses the spatial structure of the image features in the fusion process, making it difficult to determine the range of the image features perceived by the model to make judgments.

5.5 Loss Function and Optimization

As there is a slight class imbalance between the DME and non-DME samples (53.19% / 46.81%, Section 4.2), the class-weighted cross-entropy loss function (PyTorch CrossEntropyLoss with a weight tensor) is used for training the network, computed from the label counts in the observed data (scikit-learn compute_class_weight('balanced', ...)). In the case of counts in Table 3, this balance formula would lead to a mean weight of about 0.940 for the DME-positive class, and about 1.068 for the Normal/DR class, giving a slight re-weighting in comparison to a 64/36 split which would result in an average weight of about 1.079 for the DME-positive class, and about 1.036 for the Normal/DR class.

The learning rates are set to five discriminative learning rates, namely, 1×10^{-5} for learning the ConvNeXt backbone parameter, 1×10^{-7} for learning the backbone parameter of Swin (now used directly in the forward pass via the fused vision vector, as described in Section 5.2), 1×10^{-4} for learning the feature-bridge convolution, and 1×10^{-3} for learning the clinical MLP and the classification head. The lower rates are used on the pretrained backbones to maintain their pretrained representations, and the higher rates on the newly initialized modules are to speed up the adaptation. The discriminative learning-rate schedule quickly decreases for epoch 1 and decreases further at epochs 2 and 3 as shown by the training log in Figure 3 (left) for each fold of NVIDIA RTX 3060 (12 GB) with PyTorch 2.11.0 (CUDA 12.8).

6. Experimental Setup and Validation Strategy

Five-fold GroupKFold cross validation (scikit-learn) was used for the model performance estimation with the Patient_ID as grouping key. This ensures that every B-scan slice from a patient is either completely in the training set for a given fold, or completely in the validation set for a given fold, so there is no patient-level data leakage (as in previous OCT classification publications, e.g., Manikandan et al., 2023). The network is re-initalized using the initial condition used in the model trained in the previous fold, and trained for 3 epochs, with a total of ~1970-1972 mini-batch-steps per epoch. On held-out test patients, the accuracy, macro F1-score, precision and recall are calculated for those patients not seen during training, using the scikit-learn metric functions, as well as the softmax probability of the DME-positive class (AUC-ROC).

7. Results

Table 1 was generated by removing the output of the notebook that was executed and comparing it to the output log file that the notebook saves when it is run, and is the per-fold results of a five-fold GroupKFold cross validation run on the validation set. Other metrics like precision, recall etc are calculated in every fold and then the average is shown in Table 2. The training-loss curves are shown in Figure 3 and a comparison of metrics per fold from the notebook is displayed.

Table 1. Per-fold cross-validation results.

Fold	**Accuracy**	**Macro F1**	**AUC-ROC**
Fold 1	94.95%	0.9463	0.9876
Fold 2	91.83%	0.9183	0.9519
Fold 3	90.01%	0.9000	0.9812
Fold 4	93.80%	0.9374	0.9831
Fold 5	90.53%	0.9051	0.9624

Fold	Accuracy	Macro F1	AUC-ROC
Mean ± SD	**92.22% ± 1.89%**	**0.9214**	**0.9732**

Table 2. Overall performance (mean across 5 folds).

Metric (mean across 5 folds)	Value
Accuracy	92.22% (± 1.89%)
Macro F1-score	0.9214
Precision	0.9081
Recall (Sensitivity)	0.9492
AUC-ROC	0.9732

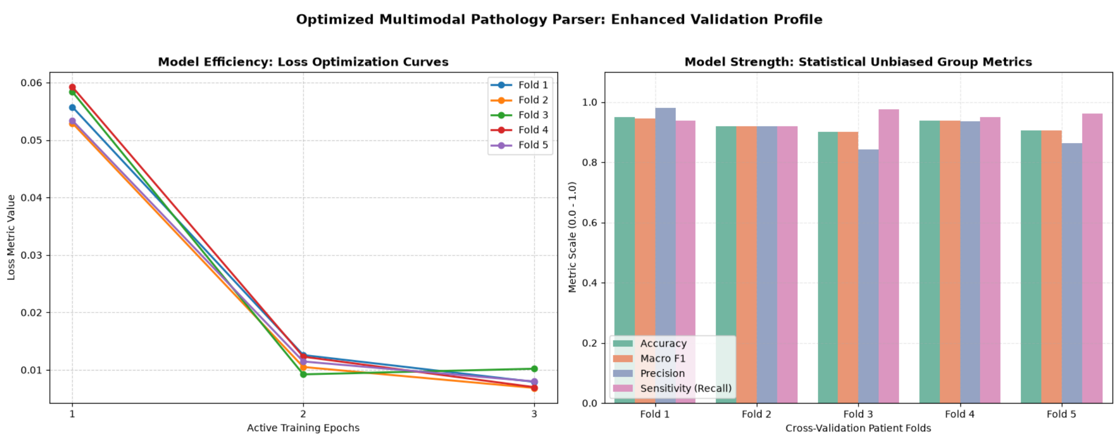


Figure 3. Left: per-fold training loss across the 3 training epochs. Right: accuracy, macro F1, precision, and recall by validation fold.

The model has the highest accuracy (94.95%) on Fold 1, and the lowest (90.01%) on Fold 3, with the AUC-ROC scores ranging from the highest (95.19%) to the lowest (98.76%) across the folds, indicating that the model seems to be more robust in its ability to rank DME-positive and DME-negative scans than in making a decision threshold. The patient split residual (or standard deviation in accuracy) across the folds is relatively low (1.89 percentage points) which is not as the result of a favorable patient split. The class-weighted loss is able to generate a higher mean of the recall (0.9492) compared to the mean of the precision (0.9081), indicating that the loss is more sensitive to the DME+ class, which is the class that is more relevant to reduce the sensitivity in a screening scenario, the DME- class. These results are achieved with the hybrid ConvNeXt–Swin backbone from Section 5.2, that is, when both vision backbones are used to generate the combined representation for classification.

8. Discussion and Validation

The results validate one of the project hypotheses: Convolutional backbone with clinical biomarkers and training with a patient-aware validation protocol leads to good discriminative performance (mean AUC-ROC 0.9732) when splitting images in classes for DME classification; and not to over optimistic statistics as image-level splitting does. Some of the design and evaluation issues discussed in the current design are not assumed to be clinically ready, but remain unresolved and are discussed below.

8.1 Fusion of the ConvNeXt and the Swin Vision Streams

The two vision streams are concatenated together or learned to pay attention to each other in a learned attention module (LAM), but the latter is not used, instead concatenated vectors of 768 dimensions are obtained. This does not learn the relative reliability of backbone to a specific region (or case), nor does it model any interaction of local features of ConvNeXt with global context of Swin other than the sum of them. It's only because of this additive-fusion approach and not a reflection that additive fusion is the best method to fuse the two backbones. All of these are natural comparisons: to compare this fusion with a concatenation (with both the 768-dimensional vectors as 1536-dimensional input for the classifier); to compare it to a small learned gating or cross-attention layer, and repeat the 5-fold cross-validation for each of these variants to obtain comparable metrics.

8.2 Interpretability

The first part of the fusion approach above is used alone: The vision stream is first averaged into a single 768 dimensional vector using adaptive average pooling and is then concatenated to the clinical embedding. This effectively makes the feature map without spatial organization and this model cannot display the disruption of the fluid-layers on a B-scan. If a fusion strategy is used for multimodal fusion, a spatial preserving fusion strategy like broadcasting the clinical features to the spatial feature map before pooling (or cross-modal attention) would be suitable to the Grad-CAM localization.

8.3 Generalization Beyond a Single Dataset

These are the results obtained from one dataset (OLIVES) from one institution. The performance of the cross validation should not be interpreted as the generalizability of the model to other sets of OCT scans such as different scanners (Topcon vs. Heidelberg), different populations, or acquisition protocols. The external validation by independent public OCT dataset would be required to make a more general claim on the generalization.

8.4 Hyperparameter and Preprocessing Justification

Some of these design decisions were made 'ad hoc': (i) clip limit in CLAHE; the size of the grid of tiles, (ii) the set of 5 discriminative learning rates, (iii) tiny-sized backbones, and (iv) hidden dimension in the clinical MLP. To understand the individual contribution of the choices of above to reported performance a pre-processing ablation (raw, CLAHE, and other contrast enhancement techniques) and a learning rate ablation would be helpful.

9. Limitations and Future Work

9.1 Limitations

- **Simple Addition Fusion Layer:** The model combines its two single models by simply adding their results which prevents the model from learning which branch is more reliable or how they interact with each other aka chance of data loss.
- **Loss of Spatial Coordinates:** The model flattens the 2D image map into a flat list of numbers before adding it with the clinical data. This causes the information loss about where the disease is in the eye.
- **Exclusion of Biomarkers:** The model wasn’t fed text info like "fluid present" to force it to actually read the raw image. This was done to test the model’s ability to find DME directly from the raw image. However, this does mean that it misses out on some valuable clinical data provided by doctors.
- **Dataset Constraints:** The AI was trained and validated using data from a single hospital’s trial. It is unknown if it will maintain its average 92% accuracy on OCT images taken from different eye machine brands or from different datasets.
- **Lack of Confidence Calibration:** The model only answers with a strict "Yes"(1) or "No"(0) regarding the disease. It does not tell us if the disease is severe or mild or different eye to eye.

9.2 Future Work

- **Better Fusion layer:** Replace the simple addition fusion layer with an "attention layer." This will allow the model to blend the image patterns with the patient's chart numbers without any loss of data.
- **Spatial Coordinates Preservation:** Inserting the patient's clinical metrics directly into the 2D image matrix before flattening it, which allows us to use tools like Grad-CAM to get visual heatmaps of the images which can show doctors exactly where the fluid is in the eye.
- **External Biomarkers:** To make the model take into account any external biomarker that can give better detection abilities.

- **External Dataset Validation:** Testing our final model parameters on entirely separate eye databases. This will prove that our architecture can work with any hospital scanner worldwide.

10. Conclusion

The goal of this project was to create a deep learning framework with a multimodal architecture for DME detection based on a ConvNeXt convolutional backbone, a Swin Transformer contextual attention stage and a lightweight clinical-feature encoder that incorporates BCVA and CST. The model yielded good results in the test: an AUC-ROC of 0.9732, a mean accuracy of 92.22% (±1.89%) and a macro F1-score of 0.9214 without any leakage of patient-level data, as previously reported in the literature for the test of classification of OCT. A check of the source code for the generation of these results revealed the Swin Transformer module is fused with the ConvNeXt features during the forward pass by element-wise addition of the two pooled vision vectors, which is the intended ConvNeXt–Swin hybrid design. The fusion approach is still a simple additive fusion that is not concatenation fusion or the learned attention-based fusion. There are some external validation and confidence calibration which are yet to be explored in the future. The baseline acquired with the ConvNeXt–Swin hybrid was excellent and well documented and the protocol for testing the project with patient groups was very well thought out and well documented with a discussion of the limitations. The patient grouped evaluation protocol is well thought out and documented with the limitations stated, and the ConvNeXt–Swin hybrid used for the project is very good and quite well documented.

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